Venkata Lakshmi Parimala Pasupuleti

Senior AI Engineer · GenAI Deployment · RAG · Agents · LLM Evaluation · Guardrails · HITL · Observability · AWS Bedrock · Python

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SUMMARY

Senior AI engineer who owns GenAI deployments end-to-end - from requirements and solution design through production launch, evaluation, and ongoing operation. Currently at Intuit integrating Anthropic Claude into production financial services: RAG pipeline with retrieval grounding, prompt and context design, structured-output schemas, LLM-as-judge evaluation harnesses with golden datasets, guardrails with sensitive-data handling, and CI/CD deployment on AWS. Built **Leasel** independently - an 8-agent LangGraph production platform with tool calling, durable state, hybrid search RAG, human-in-the-loop controls for regulated decisions, and a CI evaluation harness (93.6% precision, 1,240 test cases). Built **AgentProbe** - an open-source LLM evaluation and AI cost management platform with offline evals, LLM-as-judge scoring, CI regression gating, and cost-aware multi-provider routing across Anthropic, OpenAI, Azure OpenAI, **AWS Bedrock**, Groq, and Ollama. At Entrata (7+ years, 100K+ users), owned complex cross-functional systems end-to-end in a regulated multi-tenant SaaS platform - delivering across product, engineering, and compliance teams across 120+ sprints. Strong in systems thinking, stakeholder communication, responsible AI, and cross-functional delivery. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

GenAI Deployment · Production-Oriented AI · Systems Thinking

RAG Pipelines · Embeddings · Hybrid Search · Retrieval

AI Agents · Tool Use · LLM Orchestration · LangGraph

Prompt & Context Design · Structured Outputs

LLM Evaluation · Golden Datasets · LLM-as-Judge · CI Regression

Guardrails · Safety Controls · Responsible AI · Red-Teaming

Human-in-the-Loop (HITL) · Compliance · Regulated Decisions

Observability · Tracing · Structured Logging · OpenTelemetry

AWS Bedrock · AWS (Lambda · API Gateway · SQS · CloudWatch)

AI Cost Management · Model Selection · Multi-Provider Routing

Stakeholder Alignment · Requirements Elicitation · Acceptance Criteria

Cross-Functional Delivery · Async Documentation · Distributed Teams

Python · FastAPI · TypeScript · Docker · Kubernetes · GitHub Actions

PostgreSQL · ChromaDB · Vector Databases · Redis

EXPERIENCE

Software Engineer

Jun 2026 - Present

Intuit — Fairfax, Virginia, USA (Remote)

- **End-to-end GenAI deployment:** Own the integration of Anthropic Claude into production Python/FastAPI financial services - elicit requirements from product and engineering stakeholders, design RAG pipeline (retrieval grounding, section-aware chunking, citation-backed answers), define prompt and context schemas for consistent structured outputs, and ship through Docker + GitHub Actions CI/CD to AWS.
- **Evaluation and safety controls:** Build LLM-as-judge evaluation harnesses with golden datasets to catch hallucinations and quality drops before release; implement guardrails with input validation, sensitive-data handling, and fallbacks; establish evaluation standards and document patterns for team-wide adoption and reuse.
- Build ML data pipelines (scikit-learn, XGBoost) from ingestion through prediction APIs; develop React/TypeScript front ends; monitor production with CloudWatch structured logging.

Senior Software Engineer

Oct 2020 - Jul 2024

Entrata — Apartment & Affordable Housing SaaS (regulated multi-tenant, 100K+ monthly active users)

- **Cross-functional ownership at scale:** Delivered 20+ features end-to-end (Python, PHP, Java, JavaScript, React, PostgreSQL) across 120+ Agile sprints - eliciting requirements from product, compliance, and engineering; defining acceptance criteria; driving delivery across distributed teams. Mentored 5+ engineers; 300+ code reviews.
- Owned the HUD income-limit compliance ETL pipeline for 6 years (complex federal business rules → 180K normalized records, validation + rollback every cycle, zero compliance violations); built RabbitMQ async event-driven processing for rent updates with idempotency guarantees; PHI-compliant AWS S3 migration (100% coverage, zero incidents).
- Optimized 200+ SQL queries (30% performance gain); Docker microservices redesign (50% scalability, 30% faster deployments); Python/Bash automation eliminating 80% of manual operational workload; internal debugging tooling cutting triage time by 40%.

Junior Software Engineer

Sep 2018 - Oct 2020

Entrata India — Pune

- Built backend modules (Python, PHP, Java, PostgreSQL); delivered large-scale data integration workflows (1.6M records); led AWS S3 migration from legacy storage.

PROJECTS

Leasel - Production GenAI Deployment: Agents · RAG · HITL · Evaluation

2026 | Python, LangGraph (8-agent), FastAPI, ChromaDB, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Owned end-to-end: problem framing → solution design → production launch → ongoing evaluation. **8-agent LangGraph workflow** with tool/function calling, durable state, and conditional routing; **RAG pipeline** (OCR ingestion, section-aware chunking, hybrid search, TF-IDF + semantic reranking, citation grounding); **HITL guardrails hard-coded** - all regulated compliance decisions route to human staff; **CI eval harness** with LLM-as-judge and golden datasets: 93.6% precision, 87.9% F1 across 1,240 test cases. Three consuming surfaces share one retrieval layer.

AgentProbe - LLM Evaluation, AI Cost Management & AWS Bedrock Routing 2025 | Python, FastAPI, PostgreSQL, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- Open-source LLM evaluation and cost management platform: **offline evals** with golden datasets and LLM-as-judge; **online quality metrics** over time; **CI regression harness** gating releases on eval thresholds; **cost-aware routing** across Anthropic, OpenAI, Azure OpenAI, **AWS Bedrock**, Groq, and Ollama - routes to cheapest model meeting quality bar; **red-team-style safety scoring**; full tracing and structured logging across all agent runs.

AI on FHIR — Clinical NLP GenAI Deployment 2024 | Python, Flask, React, BERT, LLMs, FHIR REST APIs, Docker | github.com/venkatalakshmiparimala/AI_on_FHIR

- Translates English/Spanish natural-language questions into FHIR REST API calls using LLMs, BERT, and embeddings - cut clinical lookup time ~60%. HIPAA-style RBAC, OAuth-secured integrations, containerized with Docker. Defined requirements, designed solution, shipped to production.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia | GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017